# ENVS 193DS Final

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# GitHub repository

## Set Up

Created repository, added description and README

```
# read in packages
library(tidyverse)
library(here)
library(janitor)
library(DHARMa)
library(ggeffects)
library(MuMIn)
library(scales)

# read in nest observation data

nests <- read_csv(here("data", "nest_data_final.csv"))

# read in personal dinner cooking data
```

## Problem 1. Research writing

### a. Transparent statistical methods

In part 1, they used a binomial generalized linear model (aka logistic). The response is a binary outcome: the probability of American Avocet microhabitat use (it either is used or it is not). The predictor is distance to water edge, so it is continuous. This makes sense to use a binomial GLM to test whether a continuous predictor variable relates to a binary response.

In part 2, the used a chi-square test. Both the response (habitat type, 5 levels) and the predictor (bird species, 3 levels) variables are categorical. It makes sense to use chi-square because they seeing if a relationship exists between categories.

### b. Suggestions for rewriting

Part 1: The probability that American Avocet's use microhabitat depends on how far the microhabitat is from the water's edge

Part 2: There is a difference in the habitats that the three bird species (American Avocet, Forster's Tern, and Black-necked Stilt) use, so there are some species that are found in certain habitats more than others.

### c. Further analysis

An additional variable that might influence the probability of American Avocet microhabitat use is the **water depth**. This is a continuous variable, measured in centimeters with a marked pole. As with most wading birds, American Avocets primarily feed by sweeping their bills through the water. It is thus likely that a microhabitat with shallower water will be used more than one with deep water, as it is easier to feed in.

## Problem 2. Data analysis

### a. Clean and wrangle

```
# create new object using nest data frame
nests_clean <- nests |>
  # keep only the Crematogaster ant species
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # create new column with the full scientific name
  mutate(scientific_name = case_match(
    ant_species,
    "AB" ~ "Crematogaster sjostedti",
    "RRB" ~ "Crematogaster mimosae",
    "BBR" ~ "Crematogaster nigriceps"
  )) |>
  # set the level order
  mutate(scientific_name = as_factor(scientific_name),
         scientific_name = fct_relevel(scientific_name,
                                       "Crematogaster sjostedti",
                                       "Crematogaster mimosae",
                                       "Crematogaster nigriceps")) |>
  # keep the three relevant columns
  select(case_control, height_cm, scientific_name)
```

```r
# display the structure of the cleaned data frame
str(nests_clean)
```

```
tibble [270 x 3] (S3: tbl_df/tbl/data.frame)
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ scientific_name: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
```

```r
# show 10 random rows
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 3
   case_control height_cm scientific_name
          <dbl>     <dbl> <fct>
 1            0      157  Crematogaster mimosae
 2            0      169  Crematogaster nigriceps
 3            0      118  Crematogaster mimosae
 4            0      174. Crematogaster mimosae
 5            0      513  Crematogaster mimosae
 6            0      166. Crematogaster mimosae
 7            0      158  Crematogaster nigriceps
 8            0      152. Crematogaster nigriceps
 9            0      174. Crematogaster nigriceps
10            1      288. Crematogaster mimosae
```

### b. Response variable

The 1s and 0s in the "case_control" column tell if a tree had a bird nest or not. A 1 means the tree was observed to have one and the 0 meant the tree was an available tree without a nest.

### c. Purpose of study

**Tree height** is a continuous variable measured in centimeters. As the tree height increases, the probability of a nest occuring should increase too, because it physically places a nest farther up from the ground and its predators. The paper did find that the probability of a bird nesting in a tree increased with height, as described in the *Results* section, *second* paragraph.

**Acacia ant species** is a categorical variable with three levels (C. sjostedti, C. mimosae, and C. nigriceps). It makes sense that the probability of nest occurring should depend on the ant species present in a tree. Trees that are defended by more aggressive ant species are more likely to have a nest because of the protection it will receive. The paper highlights this in the *Discussion* section, *first* paragraph.

### d. Table of models

| Model number | Model description |
|---|---|
| 1 | tree height and ant species as predictors, with an interaction term |
| 2 | tree height and ant species as predictors, without an interaction term |

### e. Running the models

```
# model 1: effect of tree height and ant species on nest presence
# w/ interaction term
model1 <- glm(
  # formula: response ~ variable * variable
  case_control ~ height_cm * scientific_name,
  # use clean data frame
  data = nests_clean,
  # binomial family for response
  family = "binomial"
)

# model 2: effect of tree height and ant species on nest presence
# w/o interaction term
model2 <- glm(
  # formula: response ~ variable + interaction variable
  case_control ~ height_cm + scientific_name,
  # use clean data frame
  data = nests_clean,
  # binomial family for response
  family = "binomial"
)
```

### f. Selecting the best model

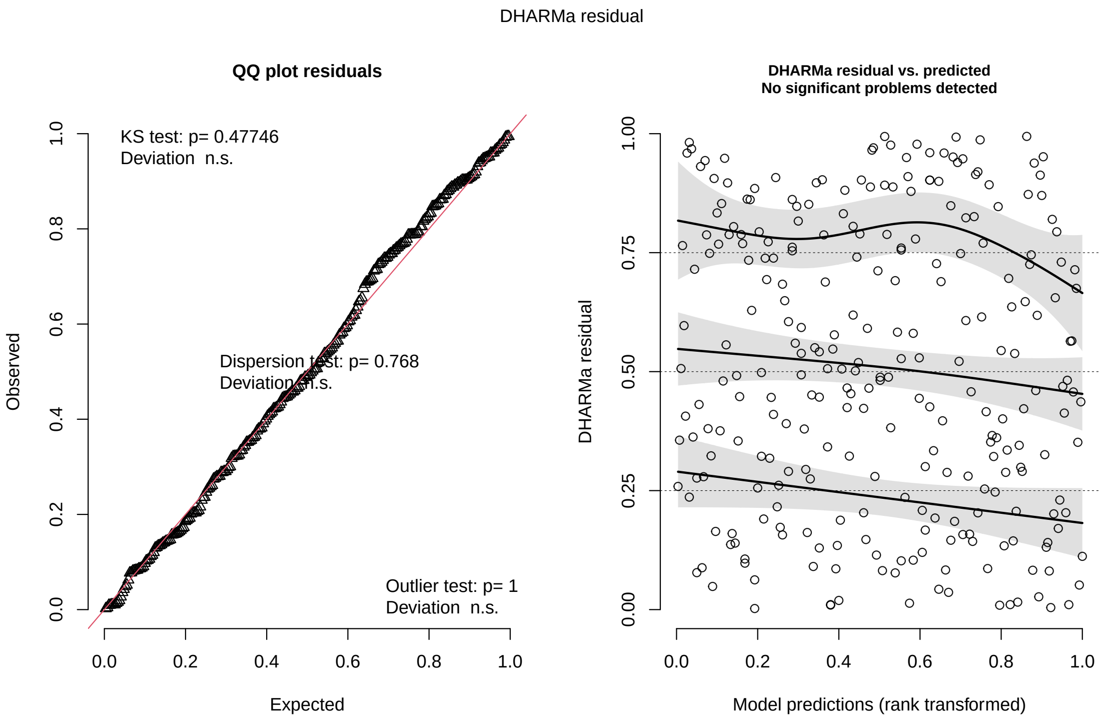
```
# compare models using AICc
AICc(model1,
     model2) |>
  arrange(AICc)
```

```
       df     AICc
model1  6 238.1236
model2  4 258.8940
```

The best model, according to Akaike's Information Criterion (AIC) uses tree height and acacia ant species as predictors of the probability of bird nest occurrence *with* an interaction.

## g. Checking the diagnostics

```
# simulate residuals and plot diagnostics
plot(simulateResiduals(model1))
```

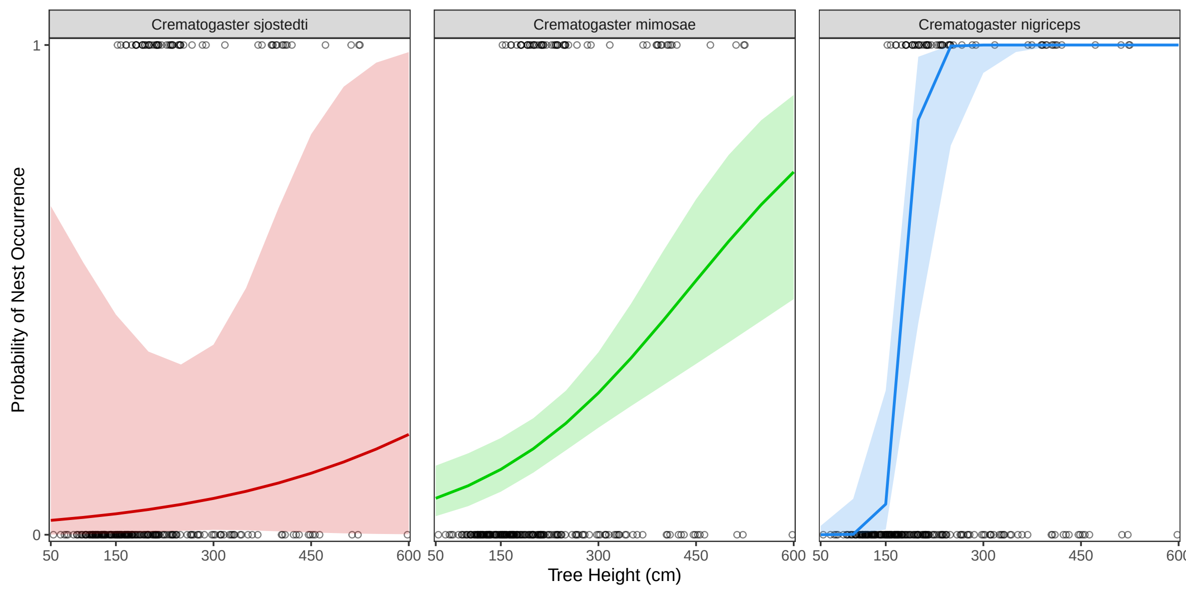


### h. Visualizing model predictions

```
# generate model predictions for height across each ant species
model_preds <- ggpredict(
  # selected model
  model1,
  # predictors
  terms = c("height_cm", "scientific_name")
)

# base layer: ggplot
ggplot(data = nests_clean,
       # tree height x-axis
       # nest occurence on y-axis
       mapping = aes(x = height_cm,
                     y = case_control)) +
  # plotting raw nest points
  geom_point(shape = 21,
             alpha = 0.5,
             col = "black") +
  # adding 95% confidence interval for ant species
  geom_ribbon(data = model_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high,
                  fill = group),
              alpha = 0.2) +
  # adding prediction lines for each ant
  geom_line(data = model_preds,
            aes(x = x,
                y = predicted,
                color = group),
            linewidth = 0.8) +
  # make y-axis to the 0-1 probability
  scale_y_continuous(limits = c(0, 1),
                     breaks = c(0, 1),
                     expand = c(0.007, 0.007)) +
  # make x-axis the range of tree heights
  scale_x_continuous(limits = c(50, 600),
                     breaks = c(50, 150, 300, 450, 600),
                     expand = c(0.005, 0.005)) +
```

```
# non-default theme
theme_bw() +
theme(panel.grid = element_blank(),
      # hide redundant legend
      legend.position = "none",
      panel.spacing.x = unit(0.5, "cm")) +
# labellinng axes
labs(x = "Tree Height (cm)",
     y = "Probability of Nest Occurrence") +
# assign line colors to each ant
scale_color_manual(values = c("Crematogaster sjostedti" = "red3",
                              "Crematogaster mimosae" = "green3",
                              "Crematogaster nigriceps" = "dodgerblue2")) +
# manually assigning ribbon colors to each ant
scale_fill_manual(values = c("Crematogaster sjostedti" = "red3",
                             "Crematogaster mimosae" = "green3",
                             "Crematogaster nigriceps" = "dodgerblue2")) +
# create separate panel for each ant
facet_wrap(~ group)
```




### i. Writing figure caption

Figure 1. The probability of bird nest occurrence increases with tree height and the strength of the relationships differs amongst acacia ant species type. Open black circles represent individual

trees, plotted by height (cm) and whether they had a bird nest (1) or no nest (0). Colored lines represent predicited probability of nest occurence from a binomial generalized linear model *with* an interaction between tree height and ant species. Shaded bands show the 95% confidence intervals about these prediction lines. Beyond colors, the separate panels distinguish each of the three ant species: Crematogaster sjostedti (red), Crematogaster mimosae (green), and Crematogaster nigriceps (blue).

Data is sourced from Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322

### j. Calculating model predictions

### k. Interpreting results

## Problem 3. Affective and exploratory visualizations

### a. Comparing visualizations

### b. Designing an analysis

### c. Checking assumptions and running analysis

### d. Creating a visualization

### e. Writing a caption

### f. Writing about results

### g. Sharing affective visualization